

For each of the following, determine whether the given three functions are linearly dependent or linearly independent on $(-\infty, \infty)$:

- (a) $y_1(t) = 1$, $y_2(t) = t$, $y_3(t) = t^2$.
 (b) $y_1(t) = -3$, $y_2(t) = 5 \sin^2 t$, $y_3(t) = \cos^2 t$.
 (c) $y_1(t) = e^t$, $y_2(t) = te^t$, $y_3(t) = t^2 e^t$.
 (d) $y_1(t) = e^t$, $y_2(t) = e^{-t}$, $y_3(t) = \cosh t$.

36. Using the definition in Problem 35, prove that if r_1, r_2 , and r_3 are distinct real numbers, then the functions $e^{r_1 t}$, $e^{r_2 t}$, and $e^{r_3 t}$ are linearly independent on $(-\infty, \infty)$. [Hint: Assume to the contrary that, say, $e^{r_1 t} = c_1 e^{r_2 t} + c_2 e^{r_3 t}$ for all t . Divide by $e^{r_2 t}$ to get $e^{(r_1 - r_2)t} = c_1 + c_2 e^{(r_3 - r_2)t}$ and then differentiate to deduce that $e^{(r_1 - r_2)t}$ and $e^{(r_3 - r_2)t}$ are linearly dependent, which is a contradiction. (Why?)]

In Problems 37–41, find three linearly independent solutions (see Problem 35) of the given third-order differential equation and write a general solution as an arbitrary linear combination of these.

37. $y''' + y'' - 6y' + 4y = 0$
 38. $y''' - 6y'' - y' + 6y = 0$
 39. $z''' + 2z'' - 4z' - 8z = 0$
 40. $y''' - 7y'' + 7y' + 15y = 0$
 41. $y''' + 3y'' - 4y' - 12y = 0$
42. (True or False): If f_1, f_2, f_3 are three functions defined on $(-\infty, \infty)$ that are pairwise linearly independent on $(-\infty, \infty)$, then f_1, f_2, f_3 form a linearly independent set on $(-\infty, \infty)$. Justify your answer.

43. Solve the initial value problem:

$$\begin{aligned} y''' - y' &= 0; & y(0) &= 2, \\ y'(0) &= 3, & y''(0) &= -1. \end{aligned}$$

44. Solve the initial value problem:

$$\begin{aligned} y''' - 2y'' - y' + 2y &= 0; \\ y(0) &= 2, & y'(0) &= 3, & y''(0) &= 5. \end{aligned}$$

45. By using Newton's method or some other numerical procedure to approximate the roots of the auxiliary equation, find general solutions to the following equations:

- (a) $3y''' + 18y'' + 13y' - 19y = 0$.
 (b) $y^{iv} - 5y'' + 5y = 0$.
 (c) $y^v - 3y^{iv} - 5y''' + 15y'' + 4y' - 12y = 0$.

46. One way to define hyperbolic functions is by means of differential equations. Consider the equation $y'' - y = 0$. The *hyperbolic cosine*, $\cosh t$, is defined as the solution of this equation subject to the initial values: $y(0) = 1$ and $y'(0) = 0$. The *hyperbolic sine*, $\sinh t$, is defined as the solution of this equation subject to the initial values: $y(0) = 0$ and $y'(0) = 1$.

- (a) Solve these initial value problems to derive explicit formulas for $\cosh t$ and $\sinh t$. Also show that $\frac{d}{dt} \cosh t = \sinh t$ and $\frac{d}{dt} \sinh t = \cosh t$.
 (b) Prove that a general solution of the equation $y'' - y = 0$ is given by $y = c_1 \cosh t + c_2 \sinh t$.
 (c) Suppose a, b , and c are given constants for which $ar^2 + br + c = 0$ has two distinct real roots. If the two roots are expressed in the form $\alpha - \beta$ and $\alpha + \beta$, show that a general solution of the equation $ay'' + by' + cy = 0$ is $y = c_1 e^{at} \cosh(\beta t) + c_2 e^{at} \sinh(\beta t)$.
 (d) Use the result of part (c) to solve the initial value problem: $y'' + y' - 6y = 0, y(0) = 2, y'(0) = -17/2$.

4.3 Auxiliary Equations with Complex Roots

The *simple harmonic equation* $y'' + y = 0$, so called because of its relation to the fundamental vibration of a musical tone, has as solutions $y_1(t) = \cos t$ and $y_2(t) = \sin t$. Notice, however, that the auxiliary equation associated with the harmonic equation is $r^2 + 1 = 0$, which has imaginary roots $r = \pm i$, where i denotes $\sqrt{-1}$.[†] In the previous section, we expressed the solutions to a linear second-order equation with constant coefficients in terms of exponential functions. It would appear, then, that one might be able to attribute a meaning to the forms e^{it} and e^{-it} and that these “functions” should be related to $\cos t$ and $\sin t$. This matchup is accomplished by Euler's formula, which is discussed in this section.

When $b^2 - 4ac < 0$, the roots of the auxiliary equation

$$(1) \quad ar^2 + br + c = 0$$

[†]Electrical engineers frequently use the symbol j to denote $\sqrt{-1}$.

associated with the homogeneous equation

$$(2) \quad ay'' + by' + cy = 0$$

are the complex conjugate numbers

$$r_1 = \alpha + i\beta \quad \text{and} \quad r_2 = \alpha - i\beta \quad (i = \sqrt{-1}),$$

where α, β are the real numbers

$$(3) \quad \alpha = -\frac{b}{2a} \quad \text{and} \quad \beta = \frac{\sqrt{4ac - b^2}}{2a}.$$

As in the previous section, we would like to assert that the functions $e^{r_1 t}$ and $e^{r_2 t}$ are solutions to the equation (2). This is in fact the case, but before we can proceed, we need to address some fundamental questions. For example, if $r_1 = \alpha + i\beta$ is a complex number, what do we mean by the expression $e^{(\alpha+i\beta)t}$? If we assume that the law of exponents applies to complex numbers, then

$$(4) \quad e^{(\alpha+i\beta)t} = e^{\alpha t + i\beta t} = e^{\alpha t} e^{i\beta t}.$$

We now need only clarify the meaning of $e^{i\beta t}$.

For this purpose, let's assume that the Maclaurin series for e^z is the same for complex numbers z as it is for real numbers. Observing that $i^2 = -1$, then for θ real we have

$$\begin{aligned} e^{i\theta} &= 1 + (i\theta) + \frac{(i\theta)^2}{2!} + \cdots + \frac{(i\theta)^n}{n!} + \cdots \\ &= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} + \cdots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots\right). \end{aligned}$$

Now recall the Maclaurin series for $\cos \theta$ and $\sin \theta$:

$$\begin{aligned} \cos \theta &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} + \cdots, \\ \sin \theta &= \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots. \end{aligned}$$

Recognizing these expansions in the proposed series for $e^{i\theta}$, we make the identification

$$(5) \quad e^{i\theta} = \cos \theta + i \sin \theta,$$

which is known as **Euler's formula**.[†]

When Euler's formula (with $\theta = \beta t$) is used in equation (4), we find

$$(6) \quad e^{(\alpha+i\beta)t} = e^{\alpha t} (\cos \beta t + i \sin \beta t),$$

which expresses the complex function $e^{(\alpha+i\beta)t}$ in terms of familiar real functions. Having made sense out of $e^{(\alpha+i\beta)t}$, we can now show (see Problem 30 on page 172) that

$$(7) \quad \frac{d}{dt} e^{(\alpha+i\beta)t} = (\alpha + i\beta) e^{(\alpha+i\beta)t},$$

[†]*Historical Footnote:* This formula first appeared in Leonhard Euler's monumental two-volume *Introductio in Analysin Infinitorum* (1748).

and, with the choices of α and β as given in (3), the complex function $e^{(\alpha+i\beta)t}$ is indeed a solution to equation (2), as is $e^{(\alpha-i\beta)t}$, and a general solution is given by

$$(8) \quad \begin{aligned} y(t) &= c_1 e^{(\alpha+i\beta)t} + c_2 e^{(\alpha-i\beta)t} \\ &= c_1 e^{\alpha t} (\cos \beta t + i \sin \beta t) + c_2 e^{\alpha t} (\cos \beta t - i \sin \beta t) . \end{aligned}$$

Example 1 shows that in general the constants c_1 and c_2 that go into (8), for a specific initial value problem, are complex.

Example 1 Use the general solution (8) to solve the initial value problem

$$y'' + 2y' + 2y = 0; \quad y(0) = 0, \quad y'(0) = 2 .$$

Solution The auxiliary equation is $r^2 + 2r + 2 = 0$, which has roots

$$r = \frac{-2 \pm \sqrt{4 - 8}}{2} = -1 \pm i .$$

Hence, with $\alpha = -1$, $\beta = 1$, a general solution is given by

$$y(t) = c_1 e^{-t} (\cos t + i \sin t) + c_2 e^{-t} (\cos t - i \sin t) .$$

For initial conditions we have

$$\begin{aligned} y(0) &= c_1 e^0 (\cos 0 + i \sin 0) + c_2 e^0 (\cos 0 - i \sin 0) = c_1 + c_2 = 0 , \\ y'(0) &= -c_1 e^0 (\cos 0 + i \sin 0) + c_1 e^0 (-\sin 0 + i \cos 0) \\ &\quad - c_2 e^0 (\cos 0 - i \sin 0) + c_2 e^0 (-\sin 0 - i \cos 0) \\ &= (-1 + i)c_1 + (-1 - i)c_2 \\ &= 2 . \end{aligned}$$

As a result, $c_1 = -i$, $c_2 = i$, and $y(t) = -ie^{-t} (\cos t + i \sin t) + ie^{-t} (\cos t - i \sin t)$, or simply $2e^{-t} \sin t$. ♦

The final form of the answer to Example 1 suggests that we should seek an alternative pair of solutions to the differential equation (2) that don't require complex arithmetic, and we now turn to that task.

In general, if $z(t)$ is a complex-valued function of the real variable t , we can write $z(t) = u(t) + iv(t)$, where $u(t)$ and $v(t)$ are real-valued functions. The derivatives of $z(t)$ are then given by

$$\frac{dz}{dt} = \frac{du}{dt} + i \frac{dv}{dt}, \quad \frac{d^2 z}{dt^2} = \frac{d^2 u}{dt^2} + i \frac{d^2 v}{dt^2} .$$

With the following lemma, we show that the complex-valued solution $e^{(\alpha+i\beta)t}$ gives rise to two linearly independent *real-valued* solutions.

Real Solutions Derived from Complex Solutions

Lemma 2. Let $z(t) = u(t) + iv(t)$ be a solution to equation (2), where a , b , and c are real numbers. Then, the real part $u(t)$ and the imaginary part $v(t)$ are real-valued solutions of (2).[†]

[†]It will be clear from the proof that this property holds for any linear homogeneous differential equation having real-valued coefficients.

Proof. By assumption, $az'' + bz' + cz = 0$, and hence

$$\begin{aligned} a(u'' + iv'') + b(u' + iv') + c(u + iv) &= 0, \\ (au'' + bu' + cu) + i(av'' + bv' + cv) &= 0. \end{aligned}$$

But a complex number is zero if and only if its real and imaginary parts are both zero. Thus, we must have

$$au'' + bu' + cu = 0 \quad \text{and} \quad av'' + bv' + cv = 0,$$

which means that both $u(t)$ and $v(t)$ are real-valued solutions of (2). ♦

When we apply Lemma 2 to the solution

$$e^{(\alpha+i\beta)t} = e^{\alpha t} \cos \beta t + ie^{\alpha t} \sin \beta t,$$

we obtain the following.

Complex Conjugate Roots

If the auxiliary equation has complex conjugate roots $\alpha \pm i\beta$, then two linearly independent solutions to (2) are

$$e^{\alpha t} \cos \beta t \quad \text{and} \quad e^{\alpha t} \sin \beta t,$$

and a general solution is

$$(9) \quad y(t) = c_1 e^{\alpha t} \cos \beta t + c_2 e^{\alpha t} \sin \beta t,$$

where c_1 and c_2 are arbitrary constants.

In the preceding discussion, we glossed over some important details concerning complex numbers and complex-valued functions. In particular, further analysis is required to justify the use of the law of exponents, Euler's formula, and even the fact that the derivative of e^{rt} is re^{rt} when r is a complex constant.[†] If you feel uneasy about our conclusions, we encourage you to substitute the expression in (9) into equation (2) to verify that it is, indeed, a solution.

You may also be wondering what would have happened if we had worked with the function $e^{(\alpha-i\beta)t}$ instead of $e^{(\alpha+i\beta)t}$. We leave it as an exercise to verify that $e^{(\alpha-i\beta)t}$ gives rise to the same general solution (9). Indeed, the sum of these two complex solutions, divided by two, gives the first real-valued solution, while their difference, divided by $2i$, gives the second.

Example 2 Find a general solution to

$$(10) \quad y'' + 2y' + 4y = 0.$$

Solution The auxiliary equation is

$$r^2 + 2r + 4 = 0,$$

which has roots

$$r = \frac{-2 \pm \sqrt{4 - 16}}{2} = \frac{-2 \pm \sqrt{-12}}{2} = -1 \pm i\sqrt{3}.$$

[†]For a detailed treatment of these topics see, for example, *Fundamentals of Complex Analysis*, 3rd ed., by E. B. Saff and A. D. Snider (Prentice Hall, Upper Saddle River, New Jersey, 2003).

Hence, with $\alpha = -1$, $\beta = \sqrt{3}$, a general solution for (10) is

$$y(t) = c_1 e^{-t} \cos(\sqrt{3}t) + c_2 e^{-t} \sin(\sqrt{3}t) . \quad \blacklozenge$$

When the auxiliary equation has complex conjugate roots, the (real) solutions oscillate between positive and negative values. This type of behavior is observed in vibrating springs.

Example 3 In Section 4.1 we discussed the mechanics of the mass–spring oscillator (Figure 4.1, page 152), and we saw how Newton’s second law implies that the position $y(t)$ of the mass m is governed by the second-order differential equation

$$(11) \quad my''(t) + by'(t) + ky(t) = 0 ,$$

where the terms are physically identified as

$$[\text{inertia}]y'' + [\text{damping}]y' + [\text{stiffness}]y = 0 .$$

Determine the equation of motion for a spring system when $m = 36$ kg, $b = 12$ kg/sec (which is equivalent to 12 N-sec/m), $k = 37$ kg/sec², $y(0) = 0.7$ m, and $y'(0) = 0.1$ m/sec. After how many seconds will the mass first cross the equilibrium point?

Solution The equation of motion is given by $y(t)$, the solution of the initial value problem for the specified values of m , b , k , $y(0)$, and $y'(0)$. That is, we seek the solution to

$$(12) \quad 36y'' + 12y' + 37y = 0 ; \quad y(0) = 0.7 , \quad y'(0) = 0.1 .$$

The auxiliary equation for (12) is

$$36r^2 + 12r + 37 = 0 ,$$

which has roots

$$r = \frac{-12 \pm \sqrt{144 - 4(36)(37)}}{72} = \frac{-12 \pm 12\sqrt{1-37}}{72} = -\frac{1}{6} \pm i .$$

Hence, with $\alpha = -1/6$, $\beta = 1$, the displacement $y(t)$ can be expressed in the form

$$(13) \quad y(t) = c_1 e^{-t/6} \cos t + c_2 e^{-t/6} \sin t .$$

We can find c_1 and c_2 by substituting $y(t)$ and $y'(t)$ into the initial conditions given in (12). Differentiating (13), we get a formula for $y'(t)$:

$$y'(t) = \left(-\frac{c_1}{6} + c_2\right)e^{-t/6} \cos t + \left(-c_1 - \frac{c_2}{6}\right)e^{-t/6} \sin t .$$

Substituting into the initial conditions now results in the system

$$\begin{aligned} c_1 &= 0.7 , \\ -\frac{c_1}{6} + c_2 &= 0.1 . \end{aligned}$$

Upon solving, we find $c_1 = 0.7$ and $c_2 = 1.3/6$. With these values, the equation of motion becomes

$$y(t) = 0.7e^{-t/6} \cos t + \frac{1.3}{6}e^{-t/6} \sin t .$$

To determine the times when the mass will cross the equilibrium point, we set $y(t) = 0$ and solve for t :

$$0 = y(t) = 0.7e^{-t/6} \cos t + \frac{1.3}{6}e^{-t/6} \sin t = (\cos t)(0.7e^{-t/6} + \frac{1.3}{6}e^{-t/6} \tan t) .$$

But $y(t)$ is not zero for $\cos t = 0$, so only the zeros of the second factor are pertinent; that is,

$$\tan t = -\frac{4.2}{1.3} .$$

A quick glance at the graph of $\tan t$ reveals that the first positive t for which this is true lies between $\pi/2$ and π . Since the arctangent function takes only values between $-\pi/2$ and $\pi/2$, the appropriate adjustment is

$$t = \arctan\left(\frac{-4.2}{1.3}\right) + \pi \approx 1.87 \text{ seconds} . \quad \blacklozenge$$

From Example 3 we see that *any* second-order constant-coefficient differential equation $ay'' + by' + cy = 0$ can be interpreted as describing a mass–spring system with mass a , damping coefficient b , spring stiffness c , and displacement y , if these constants make sense physically; that is, if a is positive and b and c are nonnegative. From the discussion in Section 4.1, then, we expect on physical grounds to see damped oscillatory solutions in such a case. This is consistent with the display in equation (9). With $a = m$ and $c = k$, the exponential decay rate α equals $-b/(2m)$, and the angular frequency β equals $\sqrt{4mk - b^2}/(2m)$, by equation (3).

It is a little surprising, then, that the solutions to the equation $y'' + 4y' + 4y = 0$ do *not* oscillate; the general solution was shown in Example 3 of Section 4.2 (page 162) to be $c_1e^{-2t} + c_2te^{-2t}$. The physical significance of this is simply that when the damping coefficient b is too high, the resulting friction prevents the mass from oscillating. Rather than overshoot the spring's equilibrium point, it merely settles in lazily. This could happen if a light mass on a weak spring were submerged in a viscous fluid.

From the above formula for the oscillation frequency β , we can see that the oscillations will not occur for $b > \sqrt{4mk}$. This *overdamping* phenomenon is discussed in more detail in Section 4.9.

It is extremely enlightening to contemplate the predictions of the mass–spring analogy when the coefficients b and c in the equation $ay'' + by' + cy = 0$ are *negative*.

Example 4 Interpret the equation

$$(14) \quad 36y'' - 12y' + 37y = 0$$

in terms of the mass–spring system.

Solution Equation (14) is a minor alteration of equation (12) in Example 3; the auxiliary equation $36r^2 - 12r + 37$ has roots $r = (+)\frac{1}{6} \pm i$. Thus, its general solution becomes

$$(15) \quad y(t) = c_1e^{+t/6} \cos t + c_2e^{+t/6} \sin t .$$

Comparing equation (14) with the mass–spring model

$$(16) \quad [\text{inertia}]y'' + [\text{damping}]y' + [\text{stiffness}]y = 0 ,$$

we have to envision a *negative* damping coefficient $b = -12$, giving rise to a friction force $F_{\text{friction}} = -by'$ that *imparts* energy to the system instead of draining it. The increase in energy over time must then reveal itself in oscillations of ever-greater amplitude—precisely in accordance with formula (15), for which a typical graph is drawn in Figure 4.7. $\blacklozenge$

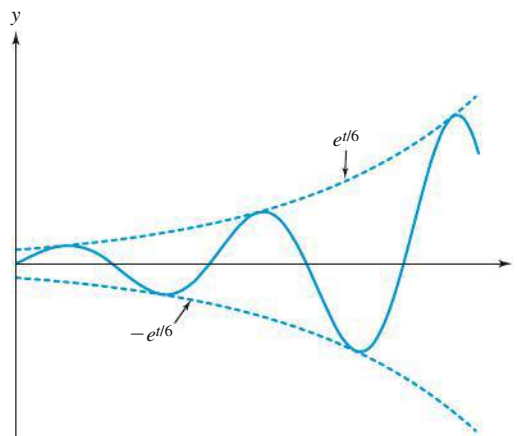


Figure 4.7 Solution graph for Example 4

Example 5 Interpret the equation

$$(17) \quad y'' + 5y' - 6y = 0$$

in terms of the mass–spring system.

Solution Comparing the given equation with (16), we have to envision a spring with a *negative* stiffness $k = -6$. What does this mean? As the mass is moved away from the spring’s equilibrium point, the spring *repels* the mass farther with a force $F_{\text{spring}} = -ky$ that intensifies as the displacement increases. Clearly the spring must “exile” the mass to (plus or minus) infinity, and we expect all solutions $y(t)$ to approach $\pm\infty$ as t increases (except for the equilibrium solution $y(t) \equiv 0$).

In fact, in Example 1 of Section 4.2, we showed the general solution to equation (17) to be

$$(18) \quad c_1 e^t + c_2 e^{-6t}.$$

Indeed, if we examine the solutions $y(t)$ that start with a unit displacement $y(0) = 1$ and velocity $y'(0) = v_0$, we find

$$(19) \quad y(t) = \frac{6 + v_0}{7} e^t + \frac{1 - v_0}{7} e^{-6t},$$

and the plots in Figure 4.8 on page 172 confirm our prediction that all (nonequilibrium) solutions diverge—except for the one with $v_0 = -6$.

What is the physical significance of this isolated bounded solution? Evidently, if the mass is given an initial inwardly directed velocity of -6 , it has barely enough energy to overcome the effect of the spring banishing it to $+\infty$ but not enough energy to cross the equilibrium point (and get pushed to $-\infty$). So it asymptotically approaches the (extremely delicate) equilibrium position $y = 0$. ♦

In Section 4.8, we will see that taking further liberties with the mass–spring interpretation enables us to predict qualitative features of more complicated equations.

Throughout this section we have assumed that the coefficients a , b , and c in the differential equation were real numbers. If we now allow them to be *complex* constants, then the roots

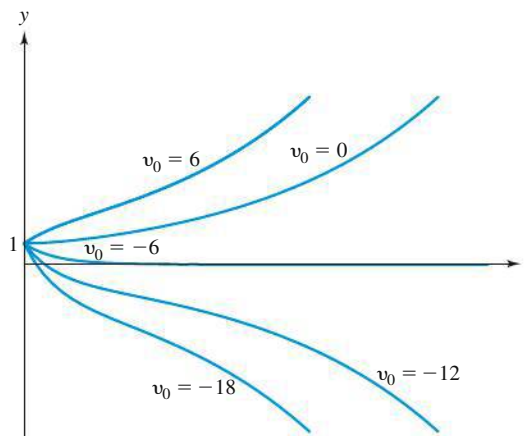


Figure 4.8 Solution graphs for Example 5

r_1, r_2 of the auxiliary equation (1) are, in general, also complex but not necessarily conjugates of each other. When $r_1 \neq r_2$, a general solution to equation (2) still has the form

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t},$$

but c_1 and c_2 are now arbitrary complex-valued constants, and we have to resort to the clumsy calculations of Example 1.

We also remark that a complex differential equation can be regarded as a system of two real differential equations since we can always work separately with its real and imaginary parts. Systems are discussed in Chapters 5 and 9.

4.3 EXERCISES

In Problems 1–8, the auxiliary equation for the given differential equation has complex roots. Find a general solution.

1. $y'' + 9y = 0$
2. $y'' + y = 0$
3. $z'' - 6z' + 10z = 0$
4. $y'' - 10y' + 26y = 0$
5. $w'' + 4w' + 6w = 0$
6. $y'' - 4y' + 7y = 0$
7. $4y'' + 4y' + 6y = 0$
8. $4y'' - 4y' + 26y = 0$

In Problems 9–20, find a general solution.

9. $y'' - 8y' + 7y = 0$
10. $y'' + 4y' + 8y = 0$
11. $z'' + 10z' + 25z = 0$
12. $u'' + 7u = 0$
13. $y'' - 2y' + 26y = 0$
14. $y'' + 2y' + 5y = 0$
15. $y'' - 3y' - 11y = 0$
16. $y'' + 10y' + 41y = 0$
17. $y'' - y' + 7y = 0$
18. $2y'' + 13y' - 7y = 0$
19. $y''' + y'' + 3y' - 5y = 0$
20. $y''' - y'' + 2y = 0$

In Problems 21–27, solve the given initial value problem.

21. $y'' + 2y' + 2y = 0$; $y(0) = 2$, $y'(0) = 1$
22. $y'' + 2y' + 17y = 0$; $y(0) = 1$, $y'(0) = -1$
23. $w'' - 4w' + 2w = 0$; $w(0) = 0$, $w'(0) = 1$

24. $y'' + 9y = 0$; $y(0) = 1$, $y'(0) = 1$
25. $y'' - 2y' + 2y = 0$; $y(\pi) = e^\pi$, $y'(\pi) = 0$
26. $y'' - 2y' + y = 0$; $y(0) = 1$, $y'(0) = -2$
27. $y''' - 4y'' + 7y' - 6y = 0$; $y(0) = 1$, $y'(0) = 0$, $y''(0) = 0$

28. To see the effect of changing the parameter b in the initial value problem

$$y'' + by' + 4y = 0; \quad y(0) = 1, \quad y'(0) = 0,$$

solve the problem for $b = 5, 4$, and 2 and sketch the solutions.

29. Find a general solution to the following higher-order equations.
 - (a) $y''' - y'' + y' + 3y = 0$
 - (b) $y''' + 2y'' + 5y' - 26y = 0$
 - (c) $y^{iv} + 13y''' + 36y'' = 0$
30. Using the representation for $e^{(\alpha+i\beta)t}$ in (6), verify the differentiation formula (7).

31. Using the mass–spring analogy, predict the behavior as $t \rightarrow +\infty$ of the solution to the given initial value problem. Then confirm your prediction by actually solving the problem.

- (a) $y'' + 16y = 0$; $y(0) = 2$, $y'(0) = 0$
 (b) $y'' + 100y' + y = 0$; $y(0) = 1$, $y'(0) = 0$
 (c) $y'' - 6y' + 8y = 0$; $y(0) = 1$, $y'(0) = 0$
 (d) $y'' + 2y' - 3y = 0$; $y(0) = -2$, $y'(0) = 0$
 (e) $y'' - y' - 6y = 0$; $y(0) = 1$, $y'(0) = 1$

32. **Vibrating Spring without Damping.** A vibrating spring without damping can be modeled by the initial value problem (11) in Example 3 by taking $b = 0$.

- (a) If $m = 10$ kg, $k = 250$ kg/sec², $y(0) = 0.3$ m, and $y'(0) = -0.1$ m/sec, find the equation of motion for this undamped vibrating spring.
 (b) After how many seconds will the mass in part (a) first cross the equilibrium point?
 (c) When the equation of motion is of the form displayed in (9), the motion is said to be **oscillatory** with **frequency** $\beta/2\pi$. Find the frequency of oscillation for the spring system of part (a).

33. **Vibrating Spring with Damping.** Using the model for a vibrating spring with damping discussed in Example 3:

- (a) Find the equation of motion for the vibrating spring with damping if $m = 10$ kg, $b = 60$ kg/sec, $k = 250$ kg/sec², $y(0) = 0.3$ m, and $y'(0) = -0.1$ m/sec.
 (b) After how many seconds will the mass in part (a) first cross the equilibrium point?
 (c) Find the frequency of oscillation for the spring system of part (a). [Hint: See the definition of frequency given in Problem 32(c).]
 (d) Compare the results of Problems 32 and 33 and determine what effect the damping has on the frequency of oscillation. What other effects does it have on the solution?

34. **RLC Series Circuit.** In the study of an electrical circuit consisting of a resistor, capacitor, inductor, and an electromotive force (see Figure 4.9), we are led to an initial value problem of the form

$$(20) \quad L \frac{dI}{dt} + RI + \frac{q}{C} = E(t);$$

$$q(0) = q_0,$$

$$I(0) = I_0,$$

where L is the inductance in henrys, R is the resistance in ohms, C is the capacitance in farads, $E(t)$ is the electromotive force in volts, $q(t)$ is the charge in coulombs on the capacitor at time t , and $I = dq/dt$ is the current in amperes. Find the current at time t if the charge on the capacitor is initially zero, the initial current is zero, $L = 10$ H, $R = 20$ Ω , $C = (6260)^{-1}$ F, and $E(t) = 100$ V. [Hint: Differentiate both sides of the

differential equation in (20) to obtain a homogeneous linear second-order equation for $I(t)$. Then use (20) to determine dI/dt at $t = 0$.]

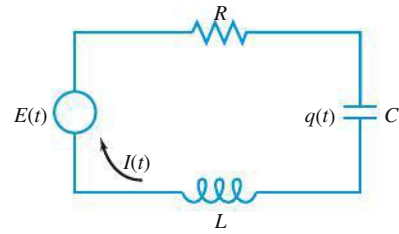


Figure 4.9 RLC series circuit

35. **Swinging Door.** The motion of a swinging door with an adjustment screw that controls the amount of friction on the hinges is governed by the initial value problem

$$I\theta'' + b\theta' + k\theta = 0; \quad \theta(0) = \theta_0, \quad \theta'(0) = v_0,$$

where θ is the angle that the door is open, I is the moment of inertia of the door about its hinges, $b > 0$ is a damping constant that varies with the amount of friction on the door, $k > 0$ is the spring constant associated with the swinging door, θ_0 is the initial angle that the door is opened, and v_0 is the initial angular velocity imparted to the door (see Figure 4.10). If I and k are fixed, determine for which values of b the door will *not* continually swing back and forth when closing.

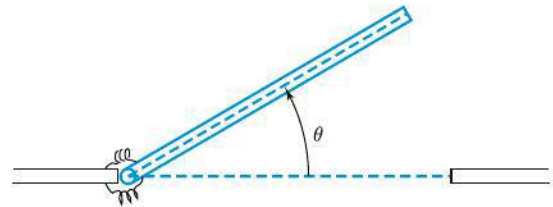


Figure 4.10 Top view of swinging door

36. Although the real general solution form (9) is convenient, it is also possible to use the form

$$(21) \quad d_1 e^{(\alpha+i\beta)t} + d_2 e^{(\alpha-i\beta)t}$$

to solve initial value problems, as illustrated in Example 1. The coefficients d_1 and d_2 are complex constants.

- (a) Use the form (21) to solve Problem 21. Verify that your form is equivalent to the one derived using (9).
 (b) Show that, in general, d_1 and d_2 in (21) must be complex conjugates in order that the solution be real.
37. The auxiliary equations for the following differential equations have repeated complex roots. Adapt the “repeated root” procedure of Section 4.2 to find their general solutions:
- (a) $y^{iv} + 2y'' + y = 0$.
 (b) $y^{iv} + 4y''' + 12y'' + 16y' + 16y = 0$. [Hint: The auxiliary equation is $(r^2 + 2r + 4)^2 = 0$.]

38. Prove the sum of angles formula for the sine function by following these steps. Fix x .

- (a) Let $f(t) := \sin(x+t)$. Show that $f''(t) + f(t) = 0$, $f(0) = \sin x$, and $f'(0) = \cos x$.
 (b) Use the auxiliary equation technique to solve the initial value problem $y'' + y = 0$, $y(0) = \sin x$, and $y'(0) = \cos x$.

- (c) By uniqueness, the solution in part (b) is the same as $f(t)$ from part (a). Write this equality; this should be the standard sum of angles formula for $\sin(x+t)$.

4.4 Nonhomogeneous Equations: the Method of Undetermined Coefficients

In this section we employ “judicious guessing” to derive a simple procedure for finding a solution to a *nonhomogeneous* linear equation with constant coefficients

$$(1) \quad ay'' + by' + cy = f(t),$$

when the nonhomogeneity $f(t)$ is a single term of a special type. Our experience in Section 4.3 indicates that (1) will have an infinite number of solutions. For the moment we are content to find one particular solution. To motivate the procedure, let's first look at a few instructive examples.

Example 1 Find a particular solution to

$$(2) \quad y'' + 3y' + 2y = 3t.$$

Solution We need to find a function $y(t)$ such that the combination $y'' + 3y' + 2y$ is a linear function of t —namely, $3t$. Now what kind of function y “ends up” as a linear function after having its zeroth, first, and second derivatives combined? One immediate answer is: *another linear function*. So we might try $y_1(t) = At$ and attempt to match up $y_1'' + 3y_1' + 2y_1$ with $3t$.

Perhaps you can see that this won't work: $y_1 = At$, $y_1' = A$ and $y_1'' = 0$ gives us

$$y_1'' + 3y_1' + 2y_1 = 3A + 2At,$$

and for this to equal $3t$, we require both that $A = 0$ and $A = 3/2$. We'll have better luck if we append a constant term to the trial function: $y_2(t) = At + B$. Then $y_2' = A$, $y_2'' = 0$, and

$$y_2'' + 3y_2' + 2y_2 = 3A + 2(At + B) = 2At + (3A + 2B),$$

which successfully matches up with $3t$ if $2A = 3$ and $3A + 2B = 0$. Solving this system gives $A = 3/2$ and $B = -9/4$. Thus, the function

$$y_2(t) = \frac{3}{2}t - \frac{9}{4}$$

is a solution to (2). ♦

Example 1 suggests the following method for finding a particular solution to the equation

$$ay'' + by' + cy = Ct^m, \quad m = 0, 1, 2, \dots;$$

namely, we guess a solution of the form

$$y_p(t) = A_mt^m + \dots + A_1t + A_0,$$

with undetermined coefficients A_j , and match the corresponding powers of t in $ay'' + by' + cy$ with Ct^m .[†] This procedure involves solving $m+1$ linear equations in the $m+1$ unknowns

[†]In this case the coefficient of t^k in $ay'' + by' + cy$ will be zero for $k \neq m$ and C for $k = m$.